

# Supporting Information

## Structure–Activity Landscape Roughness Predicts Per-Compound QSAR Error Independently of the Applicability Domain

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This document contains: per-target Spearman correlations between each roughness descriptor and per-compound random-forest error, with the applicability-domain-controlled partial correlation and the activity-cliff fraction, for all 30 benchmark targets (Table S1); the full robustness sweep across neighborhood size  $k$  and molecular distance metric (Table S2); and extended graph-network architecture and training details with per-target errors (Table S3). All values are regenerated from the analysis pipeline by `src/make_si_tables.py`.

### S1. Per-target correlations

For each target, the Spearman rank correlation  $\rho$  between each descriptor and the per-compound random-forest error is reported, computed on that target’s test set. “Cliff %” is the fraction of test compounds that participate in at least one benchmark-labeled activity cliff. “Nbr. disp. | AD” is the partial Spearman correlation of the activity-free neighborhood dispersion with error after controlling (by rank residualization) for nearest-neighbor similarity and local density. Dirichlet and Lipschitz use the query’s own activity; neighborhood dispersion and SALI density do not.

**Table S1:** Per-target Spearman correlations with per-compound random-forest error, and activity-cliff fractions, for all 30 MoleculeACE targets ( $k = 10$ , ECFP4/Tanimoto).

| Target          | $n$ | Cliff % | Dirichlet | Lipschitz | Nbr. disp. | SALI mean | Nbr. disp.   AD |
|-----------------|-----|---------|-----------|-----------|------------|-----------|-----------------|
| CHEMBL1862_Ki   | 161 | 41.6    | 0.62      | 0.38      | 0.33       | 0.23      | 0.25            |
| CHEMBL1871_Ki   | 134 | 23.9    | 0.76      | 0.51      | 0.32       | 0.21      | 0.25            |
| CHEMBL2034_Ki   | 152 | 34.2    | 0.77      | 0.65      | 0.45       | 0.34      | 0.35            |
| CHEMBL2047_EC50 | 128 | 39.1    | 0.69      | 0.51      | 0.34       | 0.28      | 0.29            |
| CHEMBL204_Ki    | 553 | 39.8    | 0.64      | 0.53      | 0.40       | 0.30      | 0.41            |
| CHEMBL2147_Ki   | 294 | 39.5    | 0.50      | 0.40      | 0.19       | 0.15      | 0.16            |
| CHEMBL214_Ki    | 666 | 36.8    | 0.65      | 0.43      | 0.27       | 0.16      | 0.22            |
| CHEMBL218_EC50  | 208 | 36.5    | 0.69      | 0.44      | 0.30       | 0.16      | 0.26            |
| CHEMBL219_Ki    | 374 | 39.6    | 0.70      | 0.50      | 0.35       | 0.15      | 0.29            |
| CHEMBL228_Ki    | 342 | 37.1    | 0.65      | 0.45      | 0.25       | 0.12      | 0.19            |
| CHEMBL231_Ki    | 197 | 24.4    | 0.67      | 0.45      | 0.32       | 0.22      | 0.21            |
| CHEMBL233_Ki    | 630 | 41.1    | 0.69      | 0.47      | 0.33       | 0.22      | 0.29            |
| CHEMBL234_Ki    | 733 | 43.7    | 0.63      | 0.43      | 0.28       | 0.13      | 0.18            |

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Table S1 (continued)

| Target          | $n$ | Cliff % | Dirichlet | Lipschitz | Nbr. disp. | SALI mean | Nbr. disp.   AD |
|-----------------|-----|---------|-----------|-----------|------------|-----------|-----------------|
| CHEMBL235_EC50  | 472 | 37.7    | 0.66      | 0.43      | 0.26       | 0.16      | 0.24            |
| CHEMBL236_Ki    | 521 | 39.2    | 0.64      | 0.40      | 0.28       | 0.21      | 0.27            |
| CHEMBL237_EC50  | 193 | 47.2    | 0.63      | 0.31      | 0.22       | 0.16      | 0.20            |
| CHEMBL237_Ki    | 522 | 42.7    | 0.67      | 0.39      | 0.27       | 0.14      | 0.26            |
| CHEMBL238_Ki    | 213 | 25.8    | 0.63      | 0.45      | 0.21       | 0.21      | 0.19            |
| CHEMBL239_EC50  | 347 | 40.6    | 0.65      | 0.42      | 0.30       | 0.16      | 0.27            |
| CHEMBL244_Ki    | 621 | 47.7    | 0.63      | 0.51      | 0.34       | 0.25      | 0.31            |
| CHEMBL262_Ki    | 173 | 18.5    | 0.63      | 0.39      | 0.12       | 0.11      | 0.13            |
| CHEMBL264_Ki    | 574 | 41.6    | 0.62      | 0.37      | 0.27       | 0.17      | 0.20            |
| CHEMBL2835_Ki   | 126 | 10.3    | 0.74      | 0.59      | 0.49       | 0.36      | 0.20            |
| CHEMBL287_Ki    | 267 | 38.6    | 0.61      | 0.38      | 0.19       | 0.01      | 0.15            |
| CHEMBL2971_Ki   | 197 | 17.3    | 0.73      | 0.52      | 0.54       | 0.32      | 0.24            |
| CHEMBL3979_EC50 | 226 | 41.6    | 0.60      | 0.41      | 0.10       | 0.02      | 0.05            |
| CHEMBL4005_Ki   | 193 | 42.0    | 0.71      | 0.63      | 0.13       | 0.14      | 0.06            |
| CHEMBL4203_Ki   | 149 | 8.7     | 0.62      | 0.61      | 0.19       | 0.11      | 0.16            |
| CHEMBL4616_EC50 | 139 | 52.5    | 0.73      | 0.48      | 0.15       | 0.02      | 0.13            |
| CHEMBL4792_Ki   | 297 | 53.9    | 0.72      | 0.47      | 0.23       | 0.16      | 0.24            |

## S2. Robustness across neighborhood size and distance metric

The central correlations were recomputed for five neighborhood sizes  $k$  under three distance metrics: Tanimoto similarity over ECFP4 (radius 2) and ECFP6 (radius 3) fingerprints, and Euclidean distance in a standardized 12-descriptor physicochemical space. *local\_var* is a uniform-weight landscape roughness (mean squared activity gap to the  $k$  neighbors, which uses the query’s activity); *nbr\_disp* is the activity-free neighborhood dispersion; *nbr\_disp | AD* is its partial correlation after controlling for nearest-neighbor distance and local density. Each cell is the median across the 30 targets. All fifteen  $k$ -metric combinations give a positive partial correlation.

## S3. Graph-network details and per-target errors

**Architecture.** The graph model is a Graph Isomorphism Network (GIN-0). Each atom carries a 16-dimensional feature vector: a one-hot atomic-number indicator over {C, N, O, F, P, S, Cl, Br, I, other}, degree (scaled by 4), formal charge, an aromaticity flag, total hydrogen count (scaled by 4), and a ring-membership flag. Three message-passing layers of hidden width 64 update node states as  $\mathbf{H}^{(l+1)} = \text{MLP}^{(l)}(\mathbf{H}^{(l)} + \mathbf{A} \mathbf{H}^{(l)})$ , where  $\mathbf{A}$  is the bond adjacency matrix and each  $\text{MLP}^{(l)}$  is a two-layer perceptron with ReLU activation. The graph embedding is the sum over layers of the masked mean of node embeddings, read out by a two-layer perceptron head.

**Training.** Models are trained with Adam (learning rate  $10^{-3}$ ) to minimize mean squared error on training-standardized targets. The *fixed* regimen runs a flat schedule of 60 epochs. The *tuned* regimen holds out 10% of the training set as a validation split, trains for up to 150 epochs with learning-rate reduction on plateau and best-checkpoint early stopping (patience 20), and uses size-bucketed mini-batches of 128 to minimize padding. Per-compound test error is  $|\hat{y}_i - y_i|$  in log-potency units. The tuned regimen lowers mean MAE on only 10 of 30 targets and does not change

**Table S2:** Median Spearman correlation (across 30 targets) with per-compound random-forest error, across neighborhood size and distance metric.

| Metric                 | $k$ | local_var | nbr_disp | nbr_disp   AD |
|------------------------|-----|-----------|----------|---------------|
| ECFP4 / Tanimoto       | 5   | 0.641     | 0.273    | 0.226         |
| ECFP4 / Tanimoto       | 10  | 0.638     | 0.275    | 0.230         |
| ECFP4 / Tanimoto       | 15  | 0.628     | 0.265    | 0.207         |
| ECFP4 / Tanimoto       | 20  | 0.615     | 0.240    | 0.188         |
| ECFP4 / Tanimoto       | 30  | 0.587     | 0.218    | 0.186         |
| ECFP6 / Tanimoto       | 5   | 0.633     | 0.264    | 0.211         |
| ECFP6 / Tanimoto       | 10  | 0.638     | 0.287    | 0.242         |
| ECFP6 / Tanimoto       | 15  | 0.623     | 0.258    | 0.210         |
| ECFP6 / Tanimoto       | 20  | 0.604     | 0.250    | 0.197         |
| ECFP6 / Tanimoto       | 30  | 0.583     | 0.219    | 0.165         |
| RDKit desc / Euclidean | 5   | 0.464     | 0.149    | 0.135         |
| RDKit desc / Euclidean | 10  | 0.482     | 0.163    | 0.143         |
| RDKit desc / Euclidean | 15  | 0.457     | 0.134    | 0.111         |
| RDKit desc / Euclidean | 20  | 0.460     | 0.138    | 0.115         |
| RDKit desc / Euclidean | 30  | 0.461     | 0.131    | 0.104         |

the roughness ordering reported in the main text.

**Table S3:** Per-target mean absolute test error (log units) for the random forest and the two graph-network regimens, with the per-target ROGI global-roughness index.

| Target          | $n$ | RF MAE | GNN (fixed-60) | GNN (tuned) | ROGI  |
|-----------------|-----|--------|----------------|-------------|-------|
| CHEMBL1862_Ki   | 161 | 0.59   | 0.78           | 0.88        | 0.049 |
| CHEMBL1871_Ki   | 134 | 0.53   | 0.61           | 0.71        | 0.056 |
| CHEMBL2034_Ki   | 152 | 0.50   | 0.59           | 0.83        | 0.074 |
| CHEMBL2047_EC50 | 128 | 0.42   | 0.64           | 0.69        | 0.078 |
| CHEMBL204_Ki    | 553 | 0.55   | 0.80           | 0.74        | 0.058 |
| CHEMBL2147_Ki   | 294 | 0.47   | 1.14           | 0.92        | 0.053 |
| CHEMBL214_Ki    | 666 | 0.51   | 0.71           | 0.69        | 0.063 |
| CHEMBL218_EC50  | 208 | 0.53   | 0.69           | 0.80        | 0.064 |
| CHEMBL219_Ki    | 374 | 0.55   | 0.77           | 0.77        | 0.073 |
| CHEMBL228_Ki    | 342 | 0.52   | 0.67           | 0.74        | 0.062 |
| CHEMBL231_Ki    | 197 | 0.56   | 0.70           | 0.68        | 0.051 |
| CHEMBL233_Ki    | 630 | 0.60   | 0.76           | 0.73        | 0.065 |
| CHEMBL234_Ki    | 733 | 0.49   | 0.66           | 0.72        | 0.065 |
| CHEMBL235_EC50  | 472 | 0.48   | 0.61           | 0.68        | 0.072 |
| CHEMBL236_Ki    | 521 | 0.52   | 0.81           | 0.69        | 0.066 |
| CHEMBL237_EC50  | 193 | 0.58   | 0.83           | 0.86        | 0.089 |
| CHEMBL237_Ki    | 522 | 0.55   | 0.70           | 0.67        | 0.068 |
| CHEMBL238_Ki    | 213 | 0.49   | 0.63           | 0.68        | 0.061 |

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Table S3 (continued)

| Target          | $n$ | RF MAE | GNN (fixed-60) | GNN (tuned) | ROGI  |
|-----------------|-----|--------|----------------|-------------|-------|
| CHEMBL239_EC50  | 347 | 0.51   | 0.68           | 0.94        | 0.085 |
| CHEMBL244_Ki    | 621 | 0.56   | 0.80           | 0.76        | 0.069 |
| CHEMBL262_Ki    | 173 | 0.57   | 0.75           | 0.76        | 0.063 |
| CHEMBL264_Ki    | 574 | 0.48   | 0.62           | 0.66        | 0.066 |
| CHEMBL2835_Ki   | 126 | 0.27   | 0.33           | 0.41        | 0.063 |
| CHEMBL287_Ki    | 267 | 0.56   | 0.66           | 0.75        | 0.071 |
| CHEMBL2971_Ki   | 197 | 0.43   | 0.58           | 0.67        | 0.043 |
| CHEMBL3979_EC50 | 226 | 0.49   | 0.66           | 0.69        | 0.076 |
| CHEMBL4005_Ki   | 193 | 0.50   | 0.66           | 0.87        | 0.072 |
| CHEMBL4203_Ki   | 149 | 0.67   | 0.79           | 0.77        | 0.052 |
| CHEMBL4616_EC50 | 139 | 0.51   | 0.65           | 0.76        | 0.110 |
| CHEMBL4792_Ki   | 297 | 0.54   | 0.68           | 0.73        | 0.096 |

## S4. Conformal prediction-interval coverage

90% split-conformal prediction intervals were built from the held-out per-compound residuals (main-text Methods). For each target the test compounds were split at random into equal calibration and evaluation halves; the nonconformity score was  $|\hat{y}_i - y_i|$  (standard) or  $|\hat{y}_i - y_i|/\sigma(x_i)$ , with  $\sigma$  an activity-free roughness (the pairwise-SALI density) or an applicability-domain risk ( $1 - s_{\text{NN}}$ ). Every scale keeps the marginal coverage guarantee; they differ only in how coverage is distributed between activity cliffs and the rest. Table S4 reports the median across the 30 targets over 50 random splits. Standard intervals under-cover cliffs; the activity-free roughness scale restores cliff coverage to the nominal level at a 15% median width cost and improves cliff coverage on 26 of 30 targets, whereas the applicability-domain scale lowers cliff coverage further.

**Table S4:** Median 90% conformal prediction-interval coverage and relative width across the 30 targets, by nonconformity scaling.

| Scaling                                      | Marginal cov. | Cliff cov. | Non-cliff cov. | Median width (rel.) |
|----------------------------------------------|---------------|------------|----------------|---------------------|
| Standard (constant width)                    | 0.904         | 0.872      | 0.920          | 1.00                |
| Roughness (SALI density)                     | 0.904         | 0.912      | 0.898          | 1.16                |
| Applicability domain ( $1 - s_{\text{NN}}$ ) | 0.904         | 0.820      | 0.954          | 1.14                |